From Labour-Supply Shock to Economic Adjustment: A Stylised Agent-Based Model of Immigration, Wages, and Firm Dynamics

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July 2026

Abstract

Empirical studies disagree on whether immigration lowers native wages because they embed different adjustment margins: some hold job counts fixed, others allow product demand, firm creation, capital, occupational specialisation, and innovation to respond. This paper develops a structurally transparent agent-based labour-market model and nests seven specifications—from a pure labour-supply benchmark (*M0*) to a full general-equilibrium model with innovation (*M6*)—to map *which assumptions* imply which signs for native employment and a native wage *index* (mean employed-native wage by skill tier, relative to pre-shock baseline; not economy-wide labour cost). Calibration uses stylised literature moments, not administrative data from any country (Borjas, 2016; Dustmann et al., 2013; OECD, 2024; Peri, 2016); simulated burn-in moments need not match those targets (Table 2). A balanced 5% labour-force shock is simulated with 50 replicates per mechanism level. Under *M0* (fixed jobs and demand), the main adjustment is quantity rationing: native employment falls to 45.2% and unemployment rises to 56.9%, while the wage index changes by only -0.05 percentage points (95% interval includes zero). Under *M6*, the wage index rises by 2.32 pp (short run) and 6.41 pp (long run) in this parameterisation, but native employment share declines over the post-shock horizon; incremental magnitudes are illustrative, not identified. Mechanism decomposition attributes 1.33 pp of short-run index change to demand (*M0*→*M1*) and 1.11 pp to firm entry (*M1*→*M2*); the large *M0*→*M1* step may reflect miscalibration. The exercise does *not* estimate a causal immigration effect for any observed economy; it nests fixed-job and general-equilibrium thought experiments in one reproducible framework.

Keywords: immigration; agent-based model; search and matching; firm dynamics; mechanism decomposition; general equilibrium
JEL codes: J15; J23; J31; J61; C63; F22

1 Introduction

Immigration expands the workforce. In the simplest textbook model, an exogenous increase in labour supply along a downward-sloping short-run demand curve reduces wages for competing workers (Borjas, 2003, 2016). Yet much of the empirical literature finds smaller, null, or even positive effects on average native wages once markets are defined broadly and adjustment is allowed time to unfold (Card, 1990; Ottaviano and Peri, 2012; Peri, 2016). The disagreement is not primarily about arithmetic; it is about *structure*: which margins—product demand, vacancy creation, capital accumulation, occupational reallocation, entrepreneurship, spatial mobility, innovation—are held fixed in the thought experiment.

Recent theoretical work formalises this distinction in search environments where immigrants may accept lower wages, be hired preferentially, and thereby induce firms to open additional vacancies (Albert, 2021). Spatial evidence shows that immigrants respond to local labour demand far more elastically than natives, attenuating geographic wage dispersion (Cadena and Kovak, 2016; Monras, 2020). Expenditure channels matter because migrants consume locally while remitting part of their income (Cheremukhin et al., 2024). Firm-level analyses show that immigration can shift entry and production organisation, particularly for high-skilled inflows (Waugh, 2018). Over longer horizons, immigration may raise innovation and productivity sufficiently to dominate short-run supply effects (Terry et al., 2026).

This paper asks:

> *Under which structural conditions does an immigration-induced labour-supply shock affect native wages and employment in the short run, and when do endogenous adjustments in demand, firm entry, capital, occupational specialisation, and innovation offset or reverse those effects?*

The contribution is methodological and diagnostic: a reproducible map from *structural assumptions* to *signs* of employment and wage-index outcomes in an artificial economy. An agent-based model (ABM) is built in which workers (natives and migrants), firms, and implicit households interact through multi-task production, search-and-matching frictions, and optional general-equilibrium feedbacks (Epstein, 2006; Railsback and Grimm, 2019). All reported magnitudes are conditional on this parameterisation; they are not estimates for any real labour market. Seven nested specifications (*M0–M6*) activate mechanisms sequentially. Counterfactual immigration shocks vary in size, skill composition, substitutability, and macroeconomic state. Outcomes are reported by skill group and time horizon (0–2, 3–7, and 8–12 years).

Parameters are chosen to reproduce stylised moments consistent with high-income OECD labour markets (Borjas, 2016; Dustmann et al., 2013; Peri, 2016), not to replicate any single country's administrative data. The model is a *computational laboratory*: an artificial economy in which mechanisms can be switched on or off to diagnose comparative statics.

Section 2 summarises evidence. Section 3 presents the model. Section 4 describes the experimental design. Section 5 reports results. Section 6 states limitations. Section 7 concludes.

2 Conceptual background and hypotheses

2.1 Why estimates diverge

Borjas (2016) catalogues how differences in skill definition, geographic market, native complementarity, and capital adjustment drive opposing conclusions. Partial-equilibrium studies that treat job counts as fixed tend to find more negative competition effects (Borjas, 2003). General-equilibrium models with product-demand feedback, heterogeneous tasks, or endogenous technology often find smaller average wage losses and sometimes gains for complementary natives (Peri, 2016; Peri and Sparber, 2009; Ottaviano and Peri, 2012).

Task-based frameworks emphasise that immigrants and natives may not compete within the same occupational cell even when they share a broad education category (Peri and Sparber, 2009). Immigrants frequently experience occupational downgrading—working below the skill level suggested by formal qualifications—which shifts competition toward specific tasks rather than entire skill tiers (Dustmann et al., 2013; Borjas, 2016).

2.2 Adjustment margins

Job creation. In search models, immigration lowers effective hiring costs when migrants accept lower reservation wages. Firms respond by posting more vacancies, creating a job-creation offset to job competition (Albert, 2021).

Demand. Immigrants earn wages and consume locally; remittances leak expenditure abroad (Cheremukhin et al., 2024).

Firm dynamics. Immigration changes factor costs and market size, shifting entry and exit (Waugh, 2018).

Innovation. Immigration can raise local innovation through diversity and scale (Terry et al., 2026).

Spatial equilibrium. Immigrants relocate toward high-demand areas, diffusing local shocks (Cadena and Kovak, 2016; Monras, 2020).

2.3 Hypotheses

- **H1. Short-run competition:** Immigration raises labour-market congestion when capital and firm numbers are fixed (*M0*); wage-index changes are small and statistically indistinguishable from zero in this parameterisation.
- **H2. Demand adjustment:** Local consumption reverses the *sign* of wage-index change relative to *M0* (*M1* vs. *M0*); magnitudes are not identified.
- **H3. Firm dynamics:** Firm entry and capital accumulation reduce labour-market congestion (*M2*, *M3*).
- **H4. Complementarity:** Natives in higher skill tiers have higher wage-index levels; inflow skill mix shifts aggregate outcomes only modestly under *M6* (*M4*).
- **H5. Entrepreneurship:** Migrant entrepreneurship accelerates job creation (*M5*).
- **H6. Dynamic reversal:** Average wage-index effects may shift from negligible under *M0* to positive in the long run when demand, firm entry, and innovation operate (*M6*).

3 Model

3.1 Overview

Figure 7 in Appendix A depicts the architecture. Time is discrete, $t = 0, 1, \ldots, T$. There are R regions, S skill sectors, and Q task types. Worker i is characterised by origin $g_i \in \{\text{native}, \text{migrant}\}$, skill s_i, occupation o_i, productivity a_i, reservation wage r_i, employment status $e_i \in \{0, 1\}$, and location z_i. Migrants additionally carry remittance rates, skill transferability, entrepreneurship propensity, and occupational downgrading risk.

Firms operate in a region–sector cell, hold capital K_j, post vacancies, and employ natives N_{jq} and migrants M_{jq} in each task q.

3.2 Production

Firm j produces with nested CES technology:

$$Y_j = A_j K_j^{\alpha} \left[\sum_{q=1}^{Q} \omega_q L_{jq}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where A_j is productivity, $\alpha \in (0,1)$ the capital share, ω_q task weights, and $\sigma > 0$ the elasticity of substitution across tasks.

Within each task, native and migrant labour combine as

$$L_{jq} = \left[\theta_q N_{jq}^{\frac{\rho_q-1}{\rho_q}} + (1-\theta_q) M_{jq}^{\frac{\rho_q-1}{\rho_q}} \right]^{\frac{\rho_q}{\rho_q-1}}. \tag{2}$$

When $\rho_q \to \infty$, natives and migrants are perfect substitutes within the task; lower ρ_q implies stronger complementarity (Peri and Sparber, 2009; Peri, 2016).

3.3 Labour market

Each period: (i) household demand; (ii) vacancy posting; (iii) separations; (iv) unemployed-to-vacancy matching; (v) production and pricing; (vi) firm entry/exit; (vii) capital adjustment; (viii) occupational mobility, innovation, and periodic relocation.

Matching follows a stylised search protocol (Pissarides, 2000): unemployed workers apply to vacancies in their region and occupation; matches form with probability increasing in the wage offer relative to reservation productivity. Immigrants' lower reservation wages imply higher job-finding rates, consistent with Albert (2021).

3.4 Outcome measures and *M0* rationing assumptions

Native wage index. The headline wage statistic is *not* economy-wide labour cost. For each skill tier s, the model records the mean wage of *employed* natives; the index change is the average across tiers of $(w_{s,t} - w_{s,t_0})/w_{s,t_0} \times 100$, where t_0 is the pre-shock year. Unemployed natives do not enter this average. Posted wage offers are bounded below and above (`wage_floor` bands), and under *M0* adjust only weakly to labour-market tightness. Readers should therefore interpret employment and unemployment as primary quantity margins; the wage index can move little even when congestion rises sharply.

***M0* fixed-margin assumptions.** When only *M0* is active: (i) firm entry, exit, and capital adjustment are disabled; (ii) local consumption demand does not feed back to sales or vacancies; (iii) vacancy targets are fixed (`adjustment_speed = fixed`); (iv) new migrants enter unemployed and compete for unchanged job slots. Under these assumptions the model implements a job-rationing thought experiment, not a competitive wage-clearing labour market. Quantity adjustment (employment/unemployment) therefore carries most of the economic content at *M0*; small wage-index changes reflect employed-only measurement and offer rigidities rather than absence of labour-market pressure.

3.5 Product demand (*M1*)

When *M1* is active, consumption feeds back to sectoral demand. Disposable income for household h satisfies

$$C_{ht} = c \max\big(Y_{ht} - R_{ht} - T_{ht},\, 0\big) + C_0, \tag{3}$$

where Y_{ht} is labour income, R_{ht} remittances (zero for natives), T_{ht} taxes, and c the marginal propensity to consume. Sectoral demand indices scale firm sales expectations, prices, and vacancy targets.

3.6 Firm dynamics (*M2*–*M3*)

Potential entrants observe expected profits $\mathbb{E}[\pi_{st}]$, entry cost F_s, and demand D_{st}:

$$P(\text{entry}_{st}) = \Lambda(\gamma_0 + \gamma_1 \mathbb{E}[\pi_{st}] - \gamma_2 F_s + \gamma_3 D_{st})\,, \tag{4}$$

where $\Lambda(x) = 1/(1 + e^{-x})$ is the logistic function. Incumbent firms exit after persistent losses; capital accumulates when utilisation and profits are high (*M3*).

3.7 Specialisation, entrepreneurship, and innovation

M4: employed natives upgrade toward their skill tier (Peri and Sparber, 2009). *M5*: unemployed migrants create firms at OECD-calibrated rates (OECD, 2024). *M6*: innovation success raises A_j with probability increasing in local skill diversity and R&D labour (Terry et al., 2026).

3.8 Immigration shock

After a 10-year burn-in, migrants arrive exogenously at $t = 10$, raising the workforce by $\Delta L/L \in \{1\%, 3\%, 5\%, 10\%\}$ with varying skill mixes. Subsequent location choices are endogenous.

4 Experimental design and calibration

4.1 Mechanism ladder

$$\begin{aligned}
&\text{M0} : \text{labour supply only (fixed firms, capital, demand)}\\
&\text{M1} : \text{M0} + \text{local consumption demand}\\
&\text{M2} : \text{M1} + \text{firm entry and exit}\\
&\text{M3} : \text{M2} + \text{capital adjustment}\\
&\text{M4} : \text{M3} + \text{task specialisation}\\
&\text{M5} : \text{M4} + \text{migrant entrepreneurship}\\
&\text{M6} : \text{M5} + \text{innovation.}
\end{aligned}$$

The incremental contribution of mechanism k is $\Delta\bar{w}^{\mathrm{M}_k} - \Delta\bar{w}^{\mathrm{M}_{k-1}}$, where $\Delta\bar{w}$ is the mean change in a native skill-wage index relative to the pre-shock baseline.

4.2 Calibration targets

Table 1 lists stylised calibration targets informed by published immigration and labour-market research. Parameters not directly identified are set by simulated-method-of-moments logic to approximate burn-in moments. Table 2 compares targets to simulated pre-shock moments under *M6*: several moments—notably native employment and the native/migrant wage ratio—deviate from the listed targets. The artificial economy is therefore stylised rather than tightly identified; counterfactual comparisons across mechanism levels remain internally valid, but levels should not be read as calibrated to any observed economy. Hold-out validation against observed historical shocks is reserved for future work (Section 6).

Table 1: Stylised calibration targets (literature-informed)

Moment	Target	Source
Native employment rate	81.5%	stylised OECD range (Borjas, 2016)
Migrant employment rate	67.0%	immigrant employment gap (Dustmann et al., 2013)
Migrant labour-force share	12%	stylised (Peri, 2016)
Native–migrant wage ratio	1.12	wage-gap literature (Dustmann et al., 2013; Borjas, 2016)
Migrant self-employment (relative)	elevated	OECD (2024)
Remittance share of migrant income	5%	literature benchmark

Table 2: Calibration targets vs. simulated pre-shock moments (*M6*, $t = 9$, mean across replicates)

Moment	Target	Simulated
Native employment rate	0.815	0.959
Migrant employment rate	0.670	0.963
Economy-wide unemployment	0.051	0.041
Native/migrant wage ratio	1.120	1.421

Note: Targets are stylised literature benchmarks (Table 1); simulated moments are from the burn-in path under the full model. Mismatch indicates the artificial economy is not tightly identified to the listed targets.

4.3 Implementation and replication

The model is implemented in Python with vectorised worker and firm arrays. Defaults: $N = 2{,}500$ workers, $R = 5$ regions, $S = 3$ skill sectors, 10-year burn-in, 12-year post-shock horizon. Each main cell uses 50 stochastic replicates; the research design envisions 500 replicates with global sensitivity analysis (Saltelli et al., 2008) as a scalability target. Documentation follows the ODD protocol (Grimm et al., 2010). Code, calibration files, result CSVs, and replication scripts are available at `https://github.com/sbf-developer/migration-labour-computational-simulation`.

5 Results

5.1 Summary

Three findings organise the simulation output; all are conditional on the assumptions in Section 3.4. First, under *M0* (fixed jobs and demand), a 5% immigration shock produces severe labour-market congestion (Table 4) while the employed-native wage index is essentially unchanged (Table 3). Second, activating demand and firm entry (*M1–M2*) reverses the *sign* of the wage index in this parameterisation; later mechanisms (*M3–M6*) contribute little at the aggregate level. Third, under *M6*, a rising wage index coexists with a gradual decline in native employment share (Figure 4); wages and employment are not interchangeable summary statistics. No finding below should be read as an estimate for a specific country.

5.2 Mechanism ladder

Table 3 and Figure 1 report native wage-index changes after a balanced 5% immigration shock. Under *M0*—labour supply only, with fixed firm count, capital, and product demand—the short-run wage index changes by -0.05 percentage points on average (standard error ≈0.07 pp; 95% interval includes zero). The main *M0* effect is not a large wage cut but severe matching congestion: native employment falls to 45.2% and economy-wide unemployment rises to 56.9% in the short run (Table 4, Figure 1b). Activating local consumption demand (*M1*) shifts the short-run wage-index effect by 1.33 percentage points and restores employment to roughly pre-shock levels—**H2** is sign-consistent, though the increment is large and may reflect demand-parameter choice rather than an identified magnitude (Section 6). Firm entry (*M1*→*M2*) contributes a further 1.11 percentage points in the short run in this run. By *M6*, the short-run wage-index effect is positive (2.32 pp) and the long-run effect (years 8–12) reaches 6.41 pp in this parameterisation; short-run native employment is 92.7% but declines over the post-shock horizon (Figure 4).

Table 3: Native wage-index change (%) by mechanism level: balanced 5% shock

Model	Short run (0–2 yr)	Long run (8–12 yr)
M0	-0.05 (0.07)	-0.34 (0.11)
M1	1.28 (0.04)	4.63 (0.06)
M2	2.39 (0.06)	6.66 (0.12)
M3	2.30 (0.06)	6.38 (0.10)
M4	2.30 (0.06)	6.38 (0.10)
M5	2.23 (0.05)	6.23 (0.10)
M6	2.32 (0.06)	6.41 (0.10)

Note: Entries are mean changes in a native wage *index* (% relative to pre-shock baseline; Section 3.4), averaged across skill tiers. This index measures employed natives' wages only; it is not a labour-cost or economy-wide wage measure. 95% replicate standard errors in parentheses. $N = 50$ replicates per cell.

Table 4: Native employment and unemployment after a balanced 5% shock

Model	Native emp. (short)	Unemp. (short)	Native emp. (long)
M0	45.2% (0.3)	56.9% (0.3)	27.2% (0.3)
M1	97.4% (0.1)	2.9% (0.1)	97.8% (0.0)
M6	92.7% (0.2)	7.9% (0.3)	86.4% (0.4)

Note: Short run = years 0–2 after shock; long run = years 8–12. Unemployment is economy-wide. *M0* reflects fixed jobs and demand (severe congestion); *M1* activates consumption demand; *M6* is the full model. 95% replicate standard errors in parentheses.

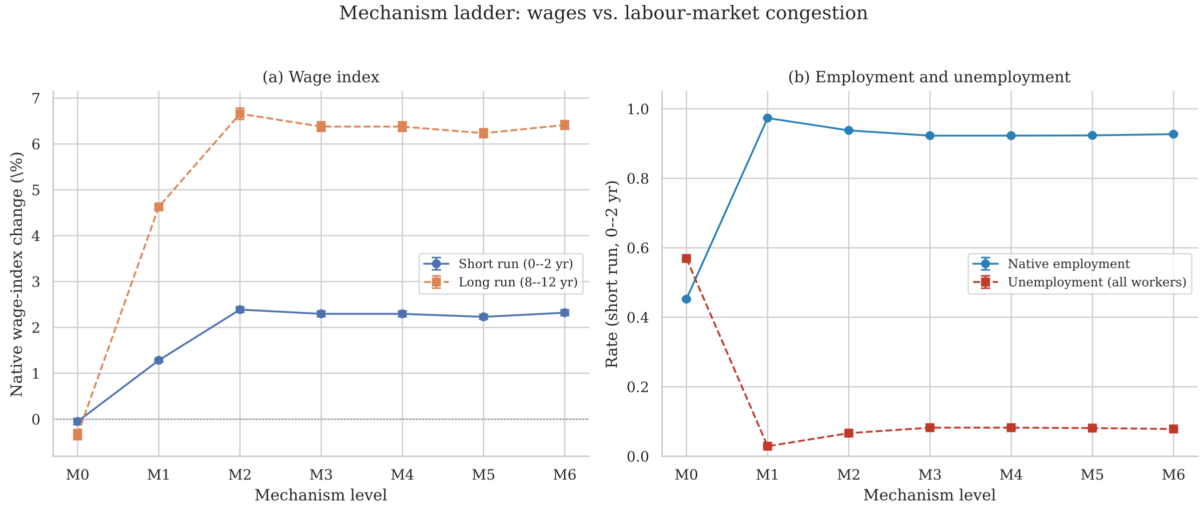

Figure 1: Mechanism ladder after a balanced 5% immigration shock at $t = 10$. Panel (a): native wage-index change (%). Panel (b): short-run native employment and economy-wide unemployment. Error bars: 95% confidence intervals across 50 replicates.

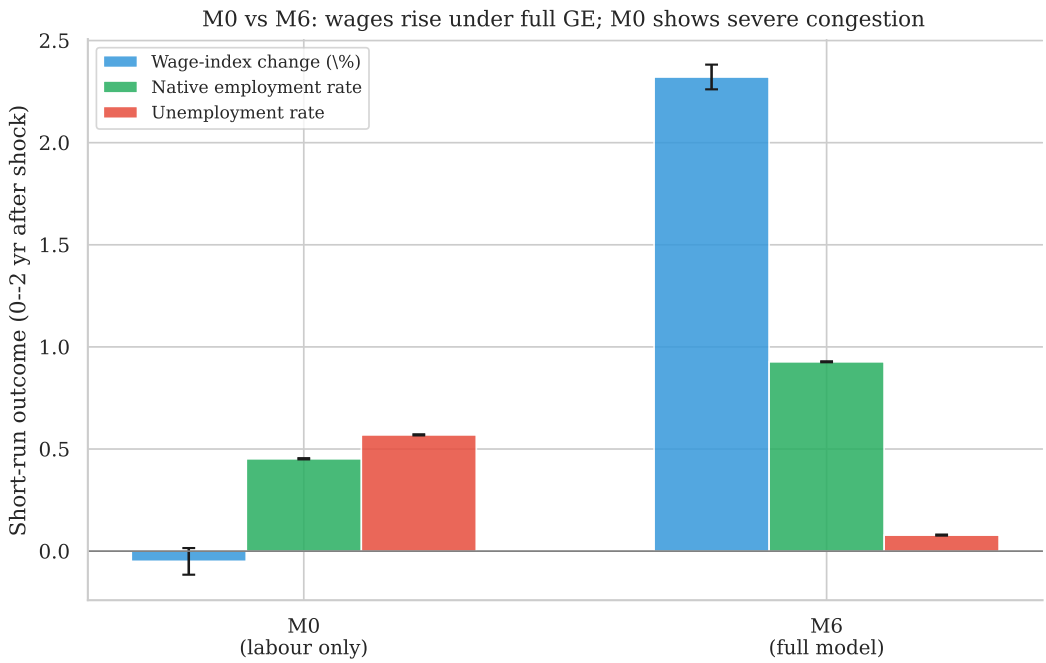

Figure 2: M0 vs. M6: short-run wage-index change, native employment, and unemployment after a 5% shock. Under fixed margins (*M0*), congestion dominates; under full adjustment (*M6*), wages rise while employment remains high in the short run but drifts down over time (Figure 4).

Figure 3 decomposes incremental mechanism contributions on the wage index. The largest single step is *M0*→*M1* (demand), followed by *M1*→*M2* (firm entry); bars for *M3*–*M6* are near zero. Because decomposition is on the employed-native index, it describes which *switches* move that statistic—not which mechanisms would dominate a labour-cost measure.

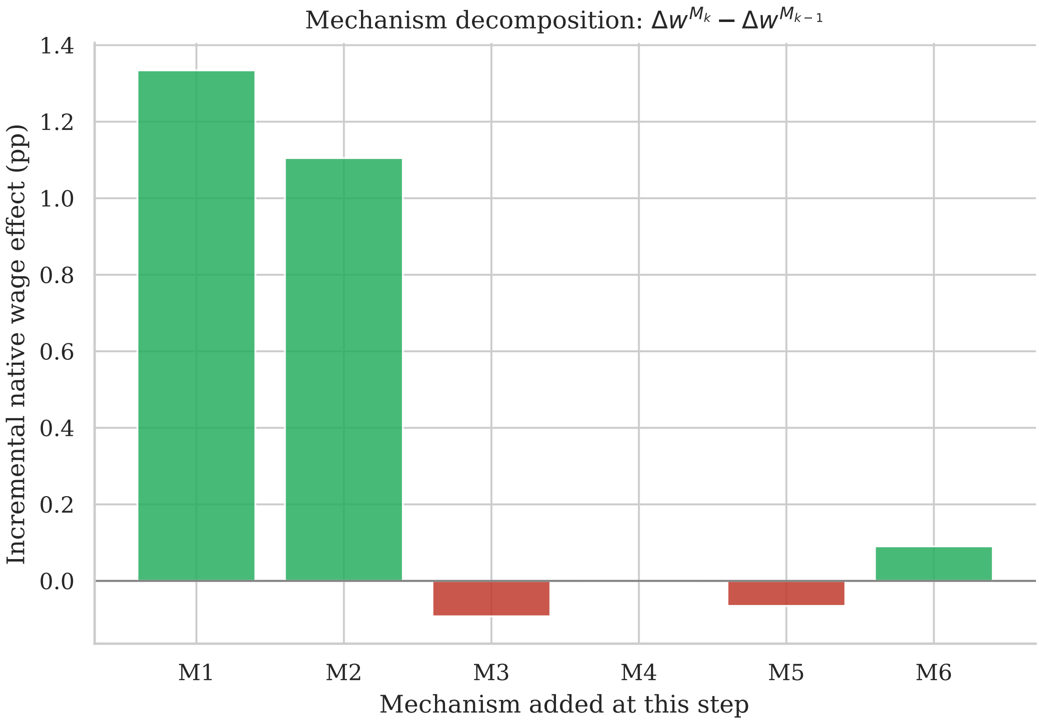

Figure 3: Mechanism decomposition: $\Delta\bar{w}^{\mathrm{M}_k} - \Delta\bar{w}^{\mathrm{M}_{k-1}}$ (percentage points, short run).

5.3 Dynamic paths

Figure 4 plots aggregate trajectories under *M6*. After the shock, the native wage index rises in this parameterisation while output per capita also increases. Native employment falls from roughly 96% pre-shock to about 86% by year 20, and unemployment rises from about 4% to 14% over the same horizon. The qualitative pattern (possible long-run wage-index gains alongside quantity adjustment) is discussed in Terry et al. (2026); no attempt is made to match any historical episode.

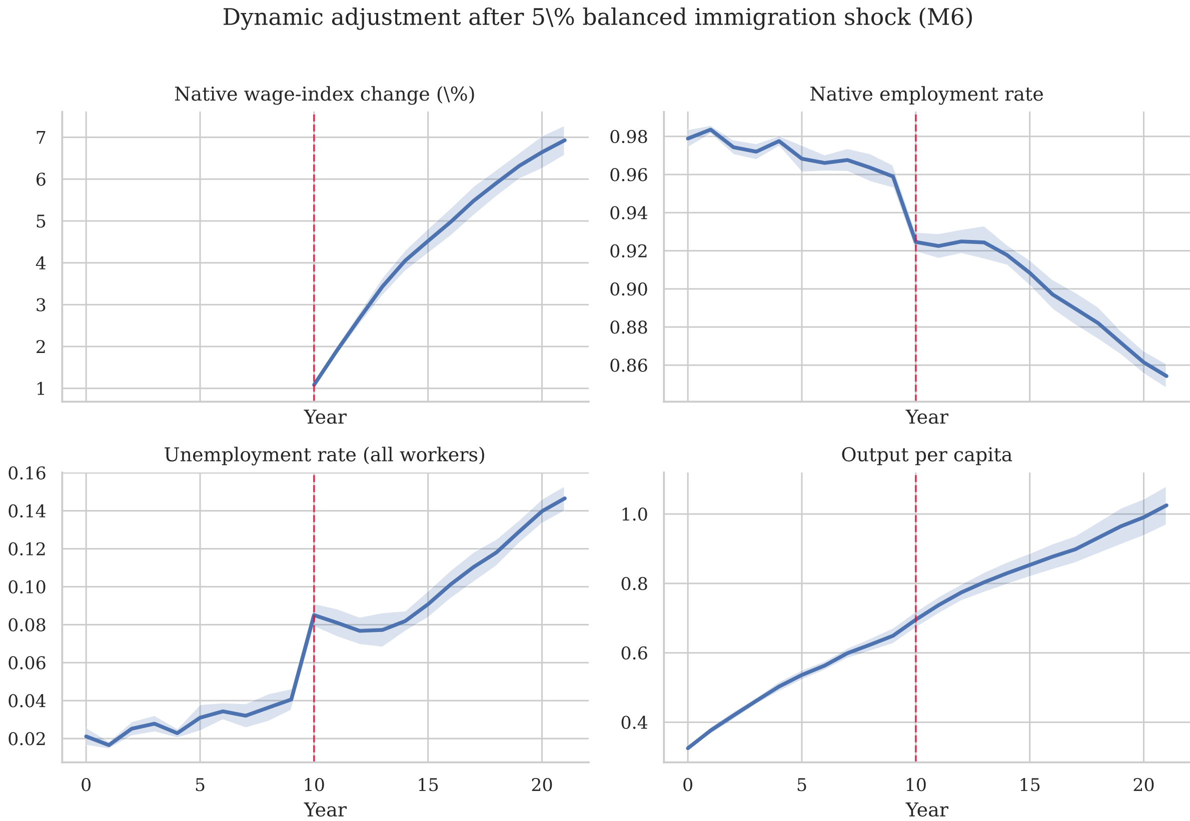

Figure 4: Dynamic adjustment under *M6* after a 5% balanced shock (vertical line at $t = 10$). Panels include unemployment; shaded bands: 95% replicate intervals.

5.4 Heterogeneity

Figure 5 varies shock size, skill composition, substitutability, and macro state under *M6* (eight replicates per scenario). All scenarios yield positive short-run wage-index changes in this parameterisation, with magnitudes between roughly 2.0 and 2.6 percentage points; unemployment rates vary only modestly across scenarios (Figure 5b). The limited spread across shock size, skill mix, and recession timing indicates weak treatment heterogeneity under the current parameterisation—not robust discrimination of policy-relevant scenarios. The clearest heterogeneity appears in the cross-section: low-skilled natives earn lower wage-index *levels* than high-skilled natives (Figure 6), consistent with stronger substitutability at the bottom of the skill distribution (Dustmann et al., 2013); the model does not report skill-specific wage *changes* in the current output.

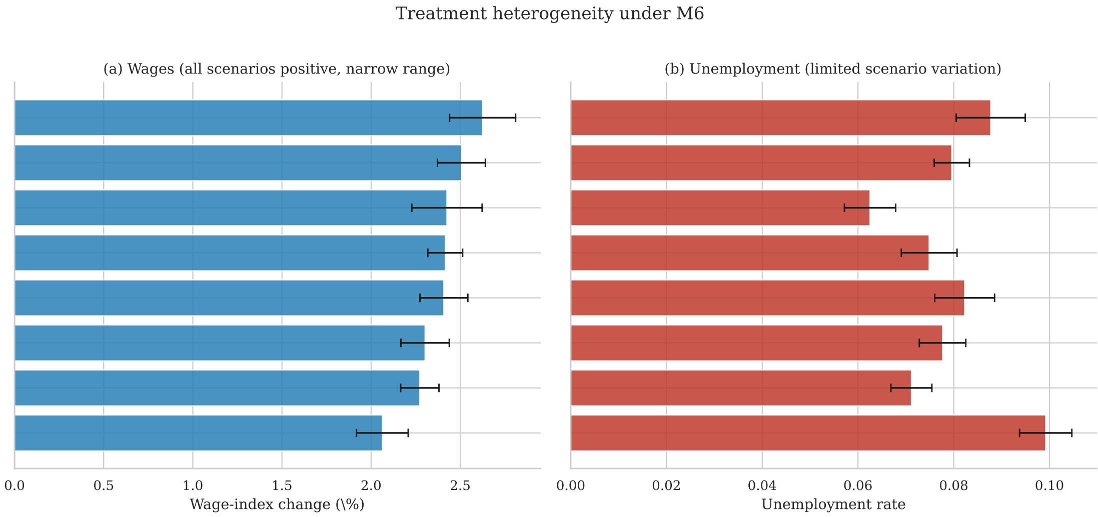

Figure 5: Treatment heterogeneity under *M6*: (a) short-run wage-index change; (b) short-run unemployment (8 replicates per scenario). Scenario variation is limited in this parameterisation.

Under *M6*, short-run native wage-index *levels* (normalised index, not percentage changes) average 0.682 for low skill and 0.898 for high skill—a cross-sectional gap consistent with stronger substitutability at the bottom of the skill distribution (Dustmann et al., 2013).

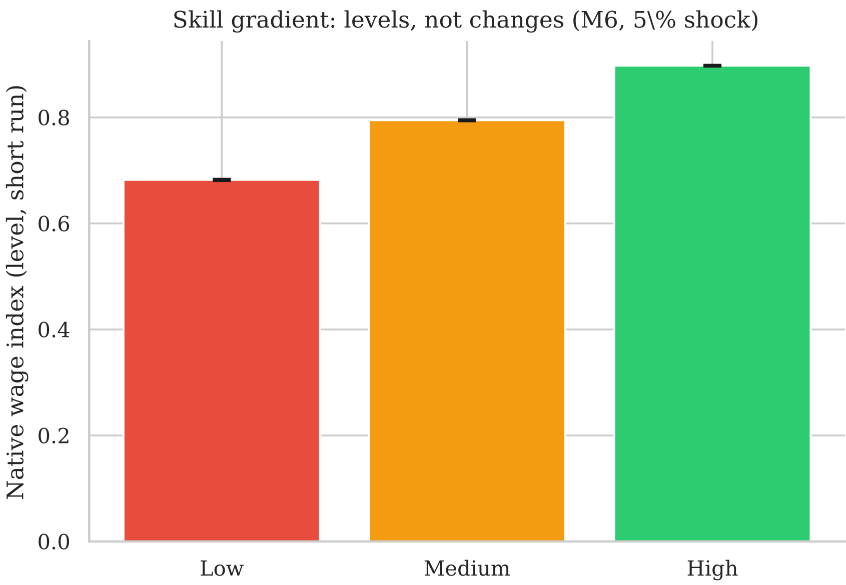

Figure 6: Native wage-index *levels* by skill tier, short run (0–2 years), *M6*, 5% shock ($N = 50$).

5.5 Hypothesis assessment

6 Limitations and scope of inference

What this study can say.

(i) In a transparent artificial economy with stylised calibration, shutting down general-equilibrium margins yields negligible short-run wage-index changes but severe unemployment under *M0*; activating demand, firm entry, and innovation can reverse the wage sign while

Table 5: Hypothesis assessment (sign and direction only; magnitudes not identified)

	Hypothesis	Assessment
H1	Competition when margins fixed	Sign-consistent (congestion)
H2	Demand reverses wage-index sign	Sign-consistent
H3	Firm/capital reduces congestion	Partial / sign only
H4	Complementary natives fare better	Partial (levels only)
H5	Migrant entrepreneurship	Not supported (≈ 0 pp)
H6	Long-run index rise under *M6*	Sign-consistent

Notes: "Sign-consistent" means the simulated direction matches the hypothesis under this parameterisation; it does *not* imply statistical identification or external validity. H1: *M0* $\approx$ -0.05 pp index, 56.9% unemployment. H2: 1.33 pp index increment at *M1* (may be miscalibrated). H4: cross-section skill gradient in levels (Figure 6); no skill-specific index changes reported. H6: -0.34 $\rightarrow$ 6.41 pp index (long run).

employment dynamics remain mixed.

(ii) Mechanism decomposition shows which *switches* move the employed-native wage index in this artificial economy; it does not identify which margins dominate in real labour markets (Borjas, 2016; Cadena and Kovak, 2016; Albert, 2021).

(iii) Distributional predictions—a persistent skill gradient in native wage levels—are stable across replicates; differential effects by inflow skill mix are small under *M6* in the current parameterisation.

What this study cannot say.

(i) It does *not* estimate the causal effect of immigration in any real economy.

(ii) The native wage index averages *employed* natives' wages by skill tier; it is not a measure of economy-wide labour cost (Section 3.4).

(iii) Positive long-run index changes under *M6* may reflect stylised demand and innovation parameters; the *M0*→*M1* step is especially large and may be miscalibrated.

(iv) Under *M6*, native employment share declines over the post-shock horizon even when the wage index rises; readers should not equate higher wages with unambiguous labour-market gains for natives.

(v) Welfare institutions, collective bargaining, language policy, and housing supply are omitted.

(vi) Task specialisation (*M4*) and migrant entrepreneurship (*M5*) contribute negligibly to the aggregate wage index in this implementation; firm entry and demand activation dominate.

(vii) The model is not validated on a hold-out historical shock with pre-registered outcomes.

(viii) Global sensitivity analysis and 500-replicate robustness are design targets not yet fully executed at scale.

Policy interpretation. Results caution against exporting partial-equilibrium wage estimates into long-run conclusions without stating which adjustment margins are inactive. They do *not* license a blanket claim that immigration raises wages in any specific country; they show that such a conclusion requires explicit assumptions about demand, firm creation, and productivity.

7 Conclusion

Immigration economics produces different answers because the underlying thought experiments differ. This paper makes those differences explicit through a nested agent-based model in an artificial economy. When only labour supply expands under fixed jobs and demand (*M0*), the model clears through unemployment rather than large wage-index changes. When general-equilibrium margins are active (*M6*), the same shock can produce a rising employed-native wage index alongside a falling native employment share in this parameterisation.

The contribution is a reproducible map from structural assumptions to signed outcomes—not a single definitive sign on "the" immigration effect and not a calibrated forecast for any country. Future work should tighten calibration, add economy-wide labour-cost metrics, validate on identified historical shocks, and conduct sensitivity analysis at scale.

Replication and code availability

Source code, calibration targets, result CSVs, and figure scripts are archived at https://github.com/sbf-developer/migration-labour-computational-simulation.

To reproduce the reported statistics and figures, run `python run_experiment.py --replicates 50` followed by `python generate_paper_outputs.py`, then compile with `pdflatex` and `bibtex` in `paper/`. All numerical values in Tables 3–5 and the abstract are auto-generated from `results/mechanism_ladder_shock_5pct_balanced.csv`.

References

Albert, C. (2021). The labor market impact of immigration: Job creation versus job competition. *American Economic Journal: Macroeconomics*, 13(1):35–78.

Borjas, G. J. (2003). The labor demand curve is downward sloping: Reexamining the impact of immigration on the labor market. *The Quarterly Journal of Economics*, 118(4):1335–1374.

Borjas, G. J. (2016). The impact of immigration: Why do studies reach such different results? *Journal of Economic Perspectives*, 30(4):31–46.

Cadena, B. C. and Kovak, B. K. (2016). Immigrants equilibrate local labor markets: Evidence from the great recession. *American Economic Journal: Applied Economics*, 8(1):257–290.

Card, D. (1990). The impact of the Mariel boatlift on the Miami labor market. *Industrial and Labor Relations Review*, 43(2):245–257.

Cheremukhin, A., Restrepo-Echavarría, A., and Tutino, A. (2024). Immigration and spatial equilibrium: The role of expenditures in U.S. markets. *American Economic Review*, 114(5):1375–1417.

Dustmann, C., Frattini, T., and Preston, I. P. (2013). The effect of immigration along the distribution of wages. *Review of Economic Studies*, 80(1):145–173.

Epstein, J. M. (2006). *Generative Social Science: Studies in Agent-Based Computational Modeling*. Princeton University Press, Princeton, NJ.

Grimm, V., Berger, U., DeAngelis, D. L., Polhill, J. G., Giske, J., and Railsback, S. F. (2010). The ODD protocol: A review and first update. *Ecological Modelling*, 221(23):2760–2768.

Monras, J. (2020). Immigration and wage dynamics: Evidence from the Mexican Peso crisis. *Journal of Political Economy*, 128(8):3017–3089.

OECD (2024). Migrant entrepreneurship in OECD countries. OECD International Migration Outlook 2024.

Ottaviano, G. I. P. and Peri, G. (2012). Rethinking the effect of immigration on wages. *Journal of the European Economic Association*, 10(1):152–197.

Peri, G. (2016). Immigrants, productivity, and labor markets. *Journal of Economic Perspectives*, 30(4):3–30.

Peri, G. and Sparber, C. (2009). Task specialization, immigration, and wages. *American Economic Journal: Applied Economics*, 1(3):135–169.

Pissarides, C. A. (2000). *Equilibrium Unemployment Theory*. MIT Press, Cambridge, MA, 2nd edition.

Railsback, S. F. and Grimm, V. (2019). *Agent-Based and Individual-Based Modeling: A Practical Introduction*. Princeton University Press, Princeton, NJ, 2nd edition.

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., and Tarantola, S. (2008). *Global Sensitivity Analysis: The Primer*. Wiley, Chichester, UK.

Terry, S. J., Chaney, T., Burchardi, K. B., Tarquinio, L., and Hassan, T. A. (2026). Immigration, innovation, and growth. *American Economic Review*, 116(3):828–861.

Waugh, M. E. (2018). Firm dynamics and immigration: The case of high-skilled immigration. In Hanson, G. H., Kerr, W. R., and Turner, S., editors, *High-Skilled Migration to the United States and Its Economic Consequences*, pages 205–238. University of Chicago Press.

A Model schematic

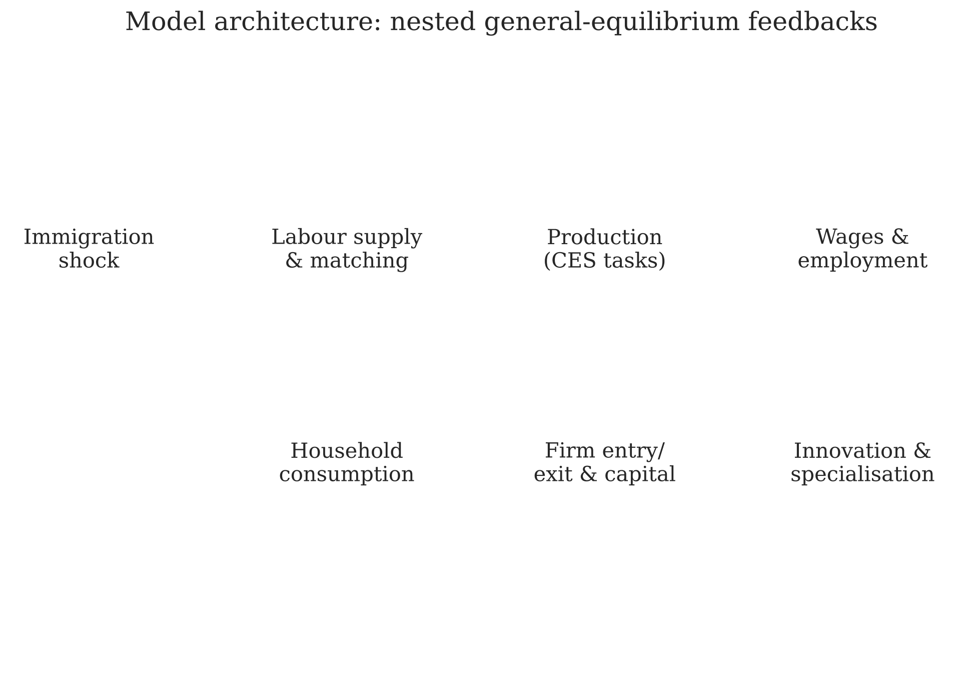

Figure 7: Model architecture: nested general-equilibrium feedbacks.

B ODD protocol summary

Purpose. Diagnose mechanism dependence of immigration wage effects.

Entities. Workers, firms, implicit households.

State variables. Worker (g, s, o, a, r, e, z); firm $(\text{sector}, \text{region}, A, K, \{N_{jq}, M_{jq}, V_{jq}, w_{jq}\}, p_j, \pi_j)$.

Process overview. Burn-in → shock at $t = 10$ → adjustment with nested mechanisms.

Design concepts. Emergence of vacancy chains; adaptation via entry and innovation; stochastic matching.

Initialization. Stylised employment and wage gaps from `data/calibration_targets.json`.

Submodels. Equations (1)–(4); search protocol in Section 3.